

2026

SMART ON-BOARD SOLUTION

DAHUA TECHNOLOGY



CONTENTS

- **COMPANY OVERVIEW**
- **BACKGROUND**
- **SCENARIO CAPABILITY**
- **SOLUTION DESIGN**
- **SUCCESS CASES**

COMPANY OVERVIEW | ABOUT DAHUA



Dahua Technology is a **world-leading video-centric smart IoT solution and service provider.**

Based on technological innovations, Dahua Technology offers end-to-end security solutions, systems, and services to create values for city operations, corporate management, and consumers.



23,000+

Employees



9700+

Patents Applications



200+

Offices in China



69

Overseas Subsidiaries and Offices



180+

Distribution Coverage
(Number of Countries)



NO.1

No.1 Global Market Share
Of Private Listed Companies

COMPANY OVERVIEW | STRATEGY

Implementing Dahua Think# 2.0 , committed to becoming the “preferred AIoT brand”

Dahua has upgraded its “Dahua Think# 2.0” corporate strategy, from intelligence to integrated intelligence. We aim to build broader AIoT capabilities through foundation models, build broader connection capabilities through multiple integration, and fully activate the value of video-centric data. This will continue to enable city and enterprise business innovation, help cities build more efficient governance systems, empower enterprises to transform and upgrade digitally, and fulfill the responsibility of "enabling a smarter society and better living."



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From Basic Monitoring to Full Autonomy

The Stage I

GPS

- Location
- Speed
- Time of driving

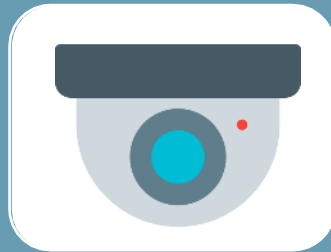


Basic Monitoring & Diagnostics

The Stage II

CCTV

- Live view
- Record down loading
- Play back



Visual Evidence & Security

The Stage III

AI Active Safety

- Active safety warning
- Driver Identification



Prevent Accidents

The Stage IV

Data-driven Safety

- Driver dashboard
- Data Analysis



Optimize Safety & Efficiency

The Stage V

Autopilot

- Autopilot



Zero Human Error



Monitoring → Security → Autonomous



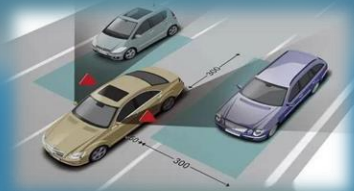
BACKGROUND | CHALLENGE & VISION

Safety



Driver Behavior

Fatigue, speeding, and sudden braking cause over 30% and 40% of accidents, posing major risks.



Blind Spot Detection

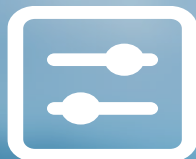
Large vehicle blind spots cause over 15% of urban accidents, requiring tech and educational fixes.



Complex road Status

Intersections, construction, and bad weather raise accident risks by 20%, needing better design, signage, and training.

Efficiency



Inefficient Manual Scheduling

Manual scheduling causes 35% empty driving and 60% cargo backlog, underscoring the need for smart systems.



Untapped Data Value

80% of companies neglect fuel and maintenance data integration, creating silos that limit data value realization.



Uneven Resource Allocation

Regional resource misallocation exceeds 40%, vehicle utilization varies widely, and manual scheduling raises costs by 15%-20%.

Revenue



Revenue loss

Driver ticket theft in some regions causes revenue loss, directly impacting bus companies' overall income



Limited revenue sources

The industry's revenue streams are limited and vulnerable to losses.



Poor Cost Control

High fuel (40%) and rising insurance (25%) costs challenge transportation profitability, requiring management optimization

VISION



Safety

Zero-Accident Goal,
Full-Link Monitoring



Efficiency

Multimodal Transport,
Intelligent Scheduling



Revenue

Green Finance,
Data Monetization

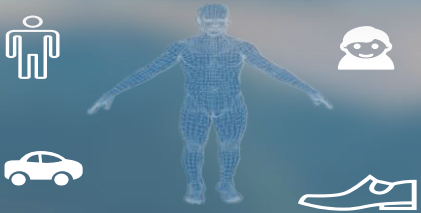
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SCENARIO CAPABILITY | AI ABILITY

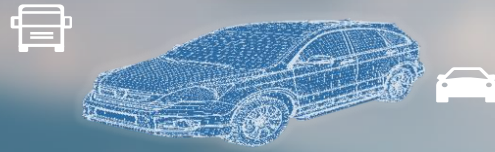


Human Body



40+ Types 8 Items

Vehicle



100+ Types 6 Items

Motorcycles



5 Types 25+ Items

Public Bus Security



Public Safety

Border Control

3 Types 25+ Items

Prison Security



Picture Quality

Video Quality

2 Types 35+ Items

Operation & Maintenance

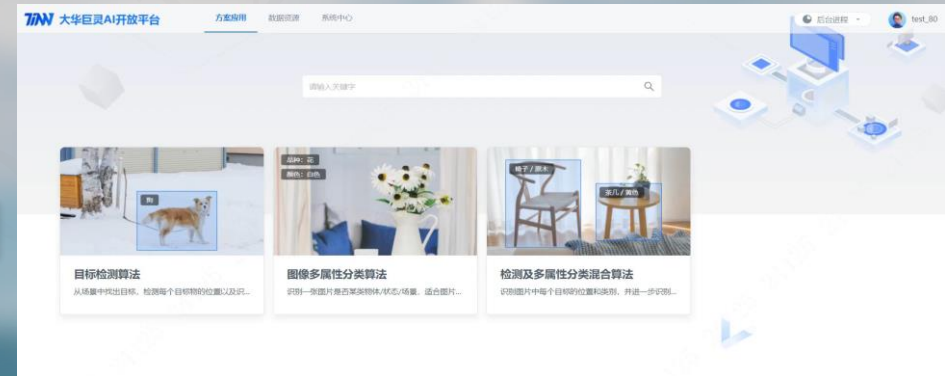


Place Safety

Audio Detection

3 Types 45+ Items

New Algorithm Training



10,000-20,000 Pic
Commercial Use

>50,000 Pic
High Accuracy

Common Algorithm

Target Detection

Video Comparison

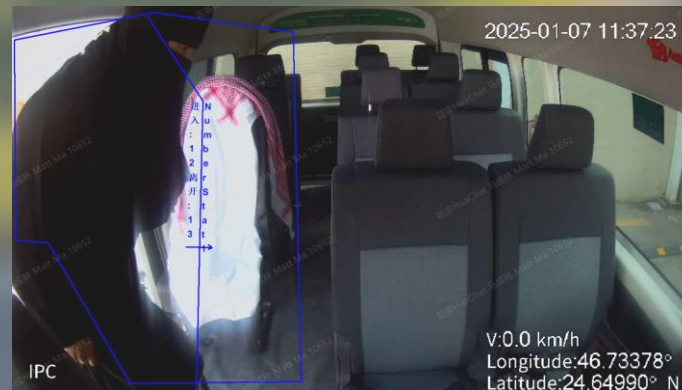
Search Comparison

Image Comparison

Algorithm Optimization for different country



Training



Special Vehicles and scenario training

Face Recognition

Face Detection

Video Structure

People Counting

General Video
Connect

AI Data Collection

Data Process

Edge AI Enable

Center AI Process

Second Level
Result Response

Multi Million
Library



Camera



AI Box



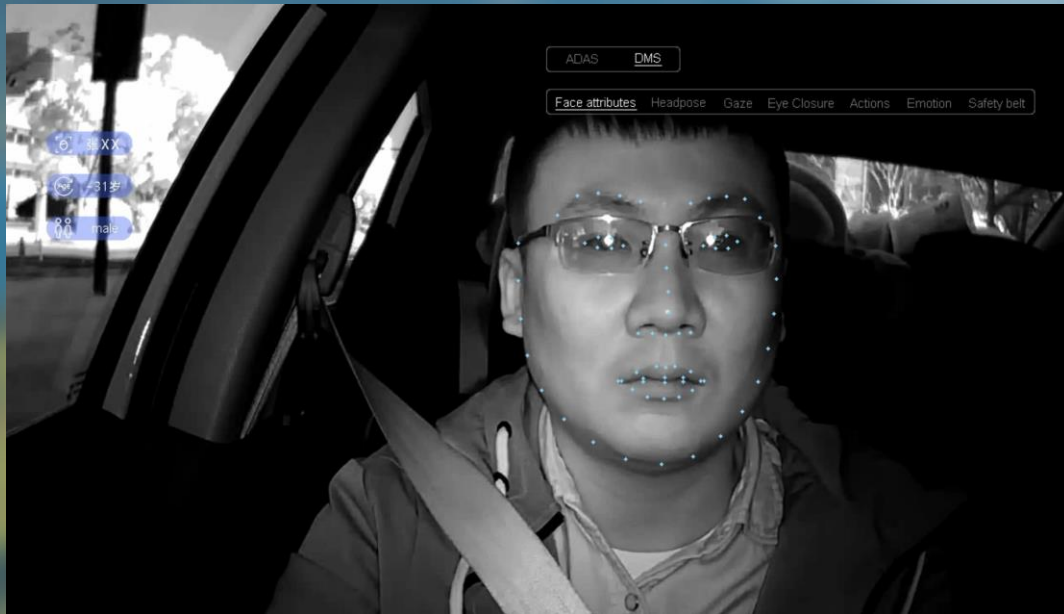
AI Server

SCENARIO CAPABILITY | ACTIVE SAFETY SYSTEM



AI-powered systems monitor driver behavior, lane position, following distance, and blind spots, issuing timely alerts to prevent accidents caused by distracted driving, tailgating, and speeding, which account for over 70% of traffic incidents.

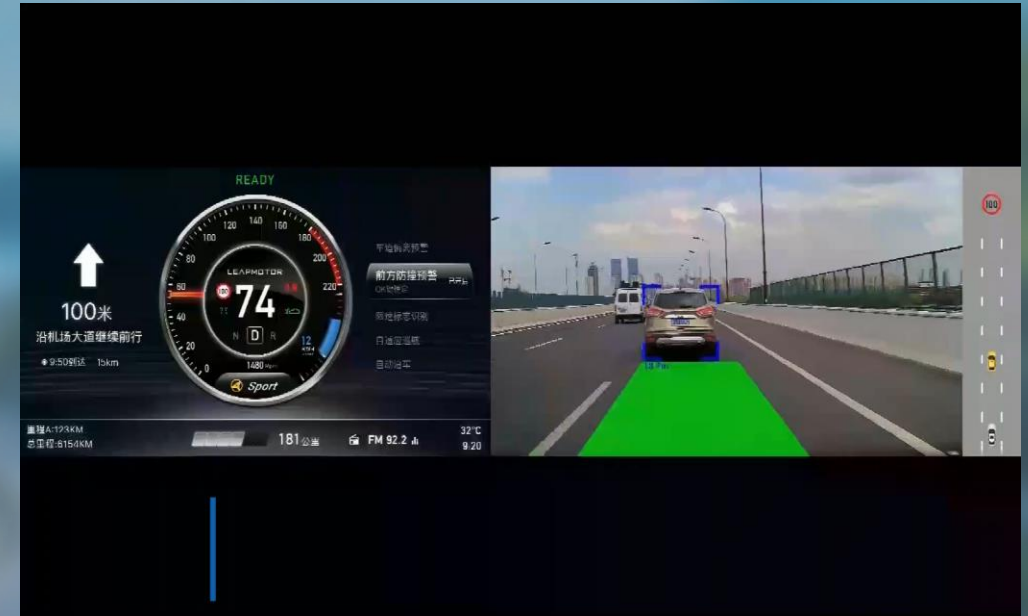
DSM



Real-Time Driver Monitoring & Safety Enhancement

The system monitors driver behavior in real time using cameras and sensors, detecting fatigue, distraction, and speeding. It issues warnings, generates reports, and aids fleet managers in enhancing safety.

ADAS

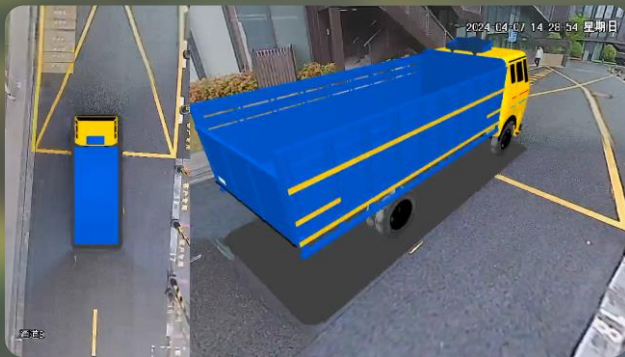
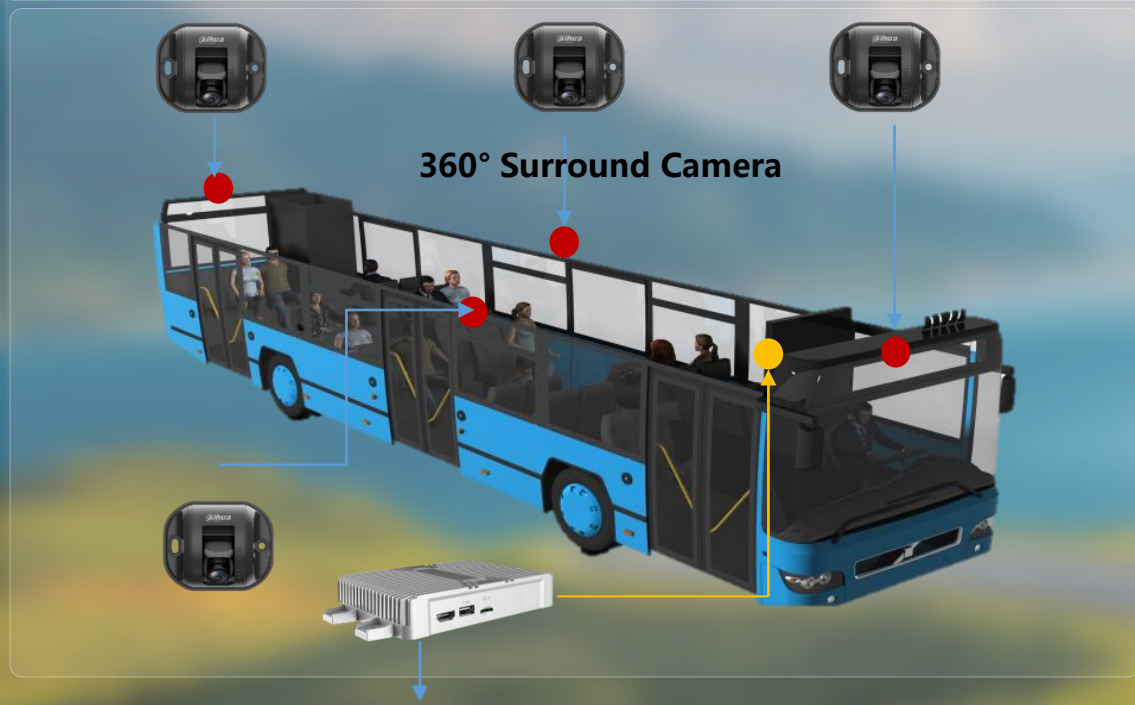


Lane, Collision, and Brake Safety Features

The ADAS system includes functions such as lane departure warning, forward collision warning, and automatic emergency braking, helping drivers avoid accidents.

SCENARIO CAPABILITY | 360° SURROUND SYSTEM

Installation Instruction



2-minute
Calibration



360°
Surround-view



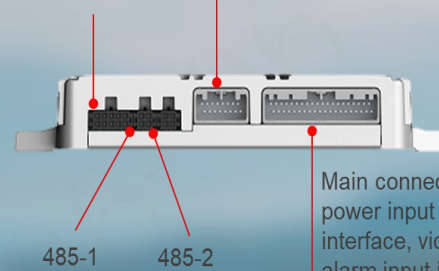
4-Way/6-Way
Video Fusion



Full Around
BSD
Detection

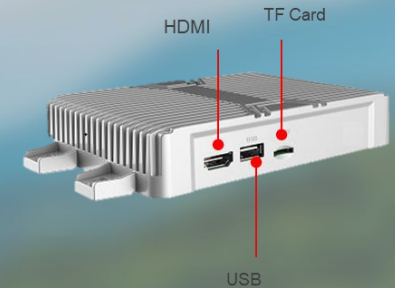
Device Parameters

EXTEND Port CH Port(5-8), for connecting analog cameras



Main connector: Mainly includes power input interface, CH interface, video output interface, alarm input interface and CAN interface.

PANA-S02-04



DAE-CVS5122-M12

- 720P resolution
- VGA, AV, and coaxial HD outputs
- DC 6-36V input
- Super capacitor for data retention during power loss



MXVR8212

- 2D display
- H.264/H.265 compression
- Two 2.5" HDDs/SSDs with anti-vibration
- Built-in GPS, 4G, and Wi-Fi



SCENARIO CAPABILITY | PASSENGER FLOW COUNTING

Installation Instruction



• IPC-HMW8441X-M/M12

Demonstration Effect



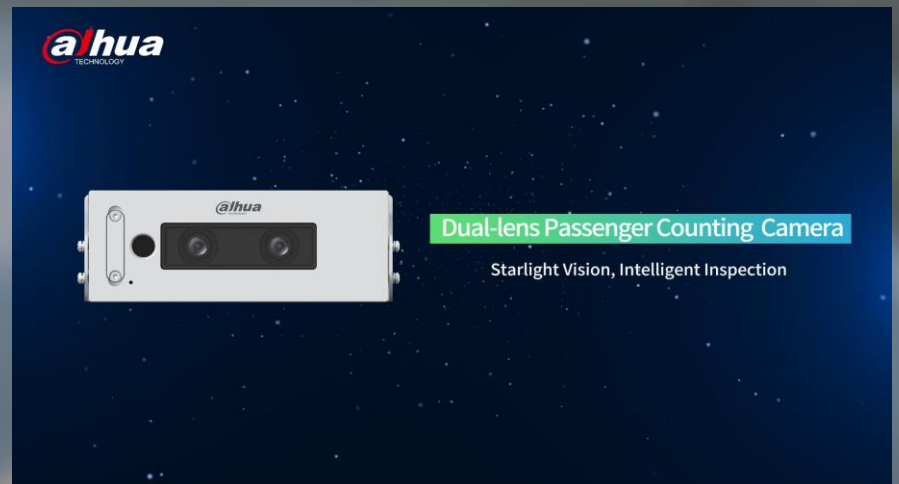
Dual-lens People Counting Camera

- Passenger flow Counting
- Support for third-party device integration
- ITxPT protocol certification
- Route scheduling, dispatch, crowd monitoring, revenue verification.

EN 45545-2



ITxPT



SCENARIO CAPABILITY | FLEET BIG DATA CENTER



Route Management and Optimization



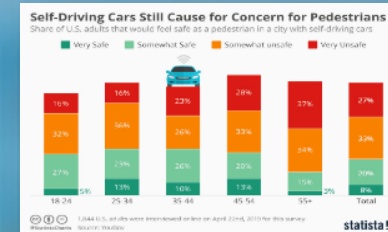
Fuel Consumption Statistics



Schedule Management and Optimization



Driver/Vehicle Safety Data



Vehicle Maintenance Data

www.ExcelDataPro.com

Vehicle Details	
Vehicle ID	Vehicle Name
1001	1001
1002	1002
1003	1003
1004	1004
1005	1005
1006	1006
1007	1007
1008	1008
1009	1009
1010	1010

Driver Identity Verification Data



Public Transport Info Inquiry



Freight Logistics Update



Student Pickup & Drop-off Info



Data-Driven Decision Making

1

Optimizes resource allocation and increases economic efficiency.

Comprehensive Safety Monitoring

2

Reduces incident rates and promotes a culture of safety among drivers.

Enhanced Transparency & Customer Experience

3

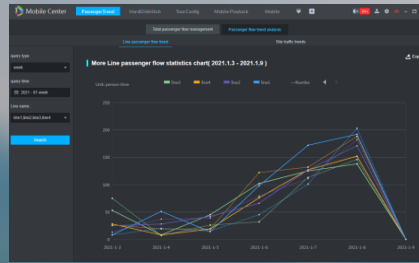
Enhances trust and reliability in public transportation services.

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- **SUCCESS CASES**

ON-BOARD SOLUTION | SMART MOBILE SOLUTION

Application



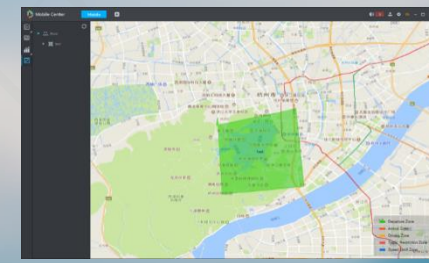
Passenger flow trend



Alarm statistics



Active safety alarm

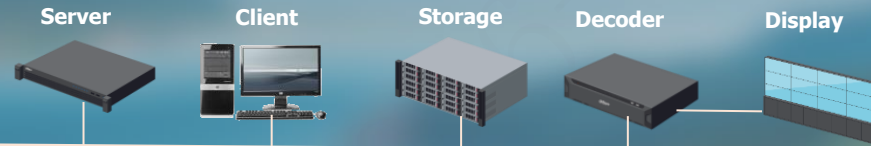


E-fence & Route management



APP

Control Center



4G



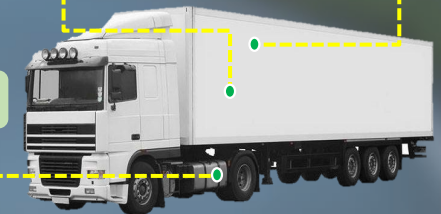
Bus



School Bus



Truck



ON-BOARD SOLUTION | SMART BUS SOLUTION

REDUCE OPERATING COSTS

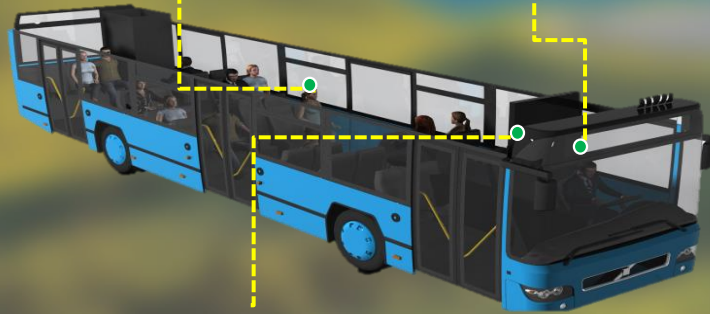
Prevents bus losses by detecting fare evasion and driver theft, via passenger counting, fare system integration, and enforcement measures.



Passengers do not pay



Driver steals ticket fare



Passenger Counting



Passenger Counting Camera



Dual-Lens Camera



Integrate with the bus fare system or conduct manual checks to calculate the total number of passengers and the corresponding bus fare, then compare it with the actual collected fares to check for discrepancies.



Punish driver



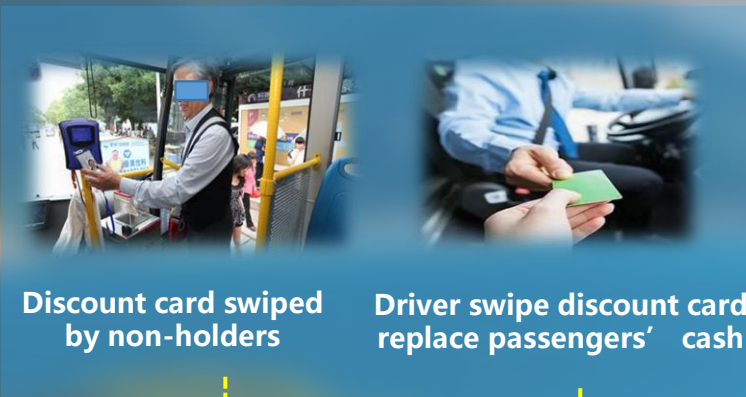
Remind passengers to buy tickets (through the public address system or by the driver)

ON-BOARD SOLUTION | SMART BUS SOLUTION



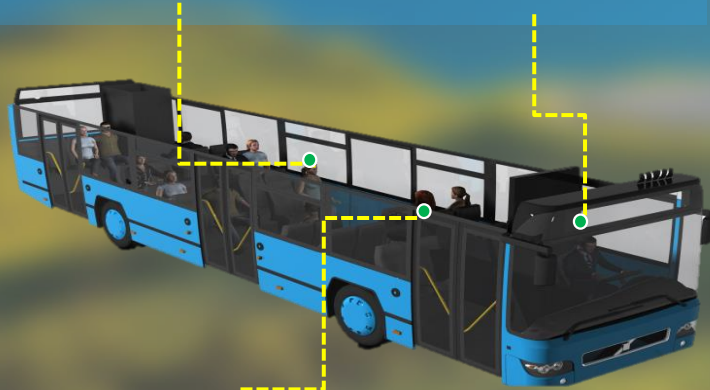
REDUCE BUS CARD FRAUD

Preventing Bus Revenue Losses from Illegal Discount Card Usage



Discount card swiped by non-holders

Driver swipe discount card replace passengers' cash



Face Recognition



Face Recognition Camera



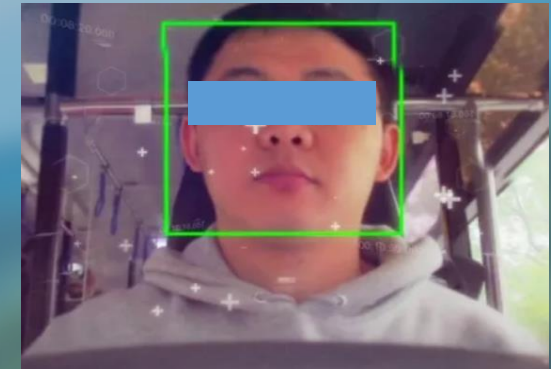
Duel-Lens passenger counting Camera



Integrate with bus card validator

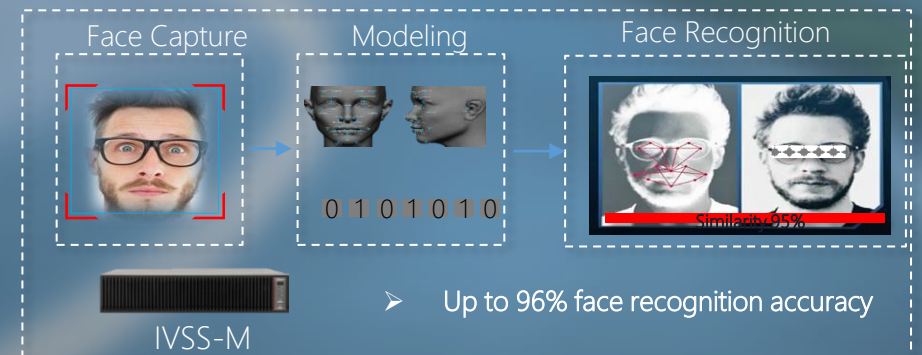


Remind the driver to check or freeze the discount card



Snapshot when discount card swiped

4G Upload to Monitor Center



If not the same person as the registered one, trigger alarm

ON-BOARD SOLUTION | SMART BUS SOLUTION

INCREASE THE REVENUE

Providing passengers with comprehensive travel guidance like next bus info, route details and fare information at bus stops and on buses, while also enabling advertising display, enhancing both the travel experience and potential revenue streams.

Guidance

Next inbound bus



- ✓ Next inbound bus
- ✓ Route list display and information(all stops,current stop, fare,last station,first/last time,location of the two nearest buses, arrival time of the nearest bus)
- ✓ Display of advertising
- ✓ Date and time, weather and temperature

Advertisement



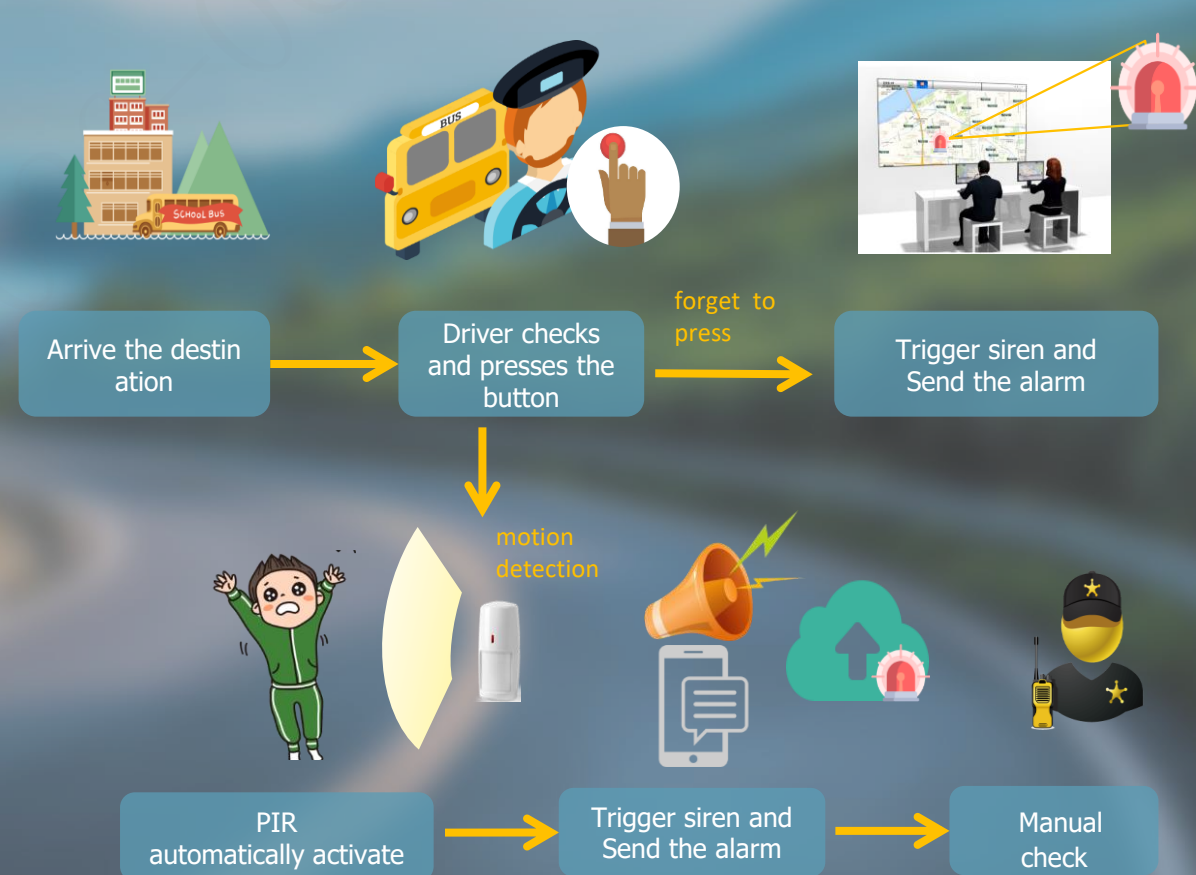
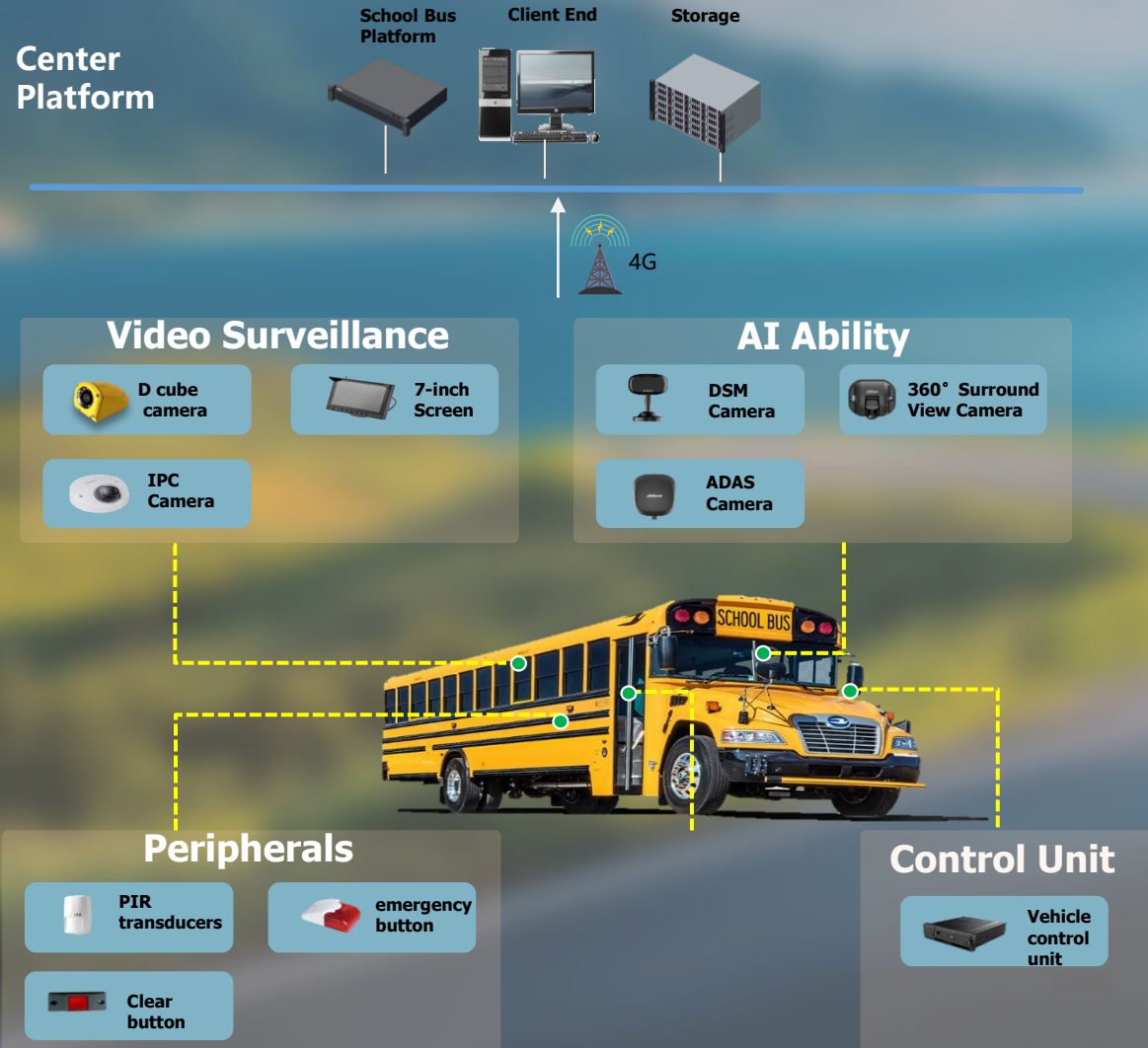
- ✓ Bus routes
- ✓ List of stations
- ✓ Starting and ending station
- ✓ First and last bus time
- ✓ Fare information
- ✓ Current station display

ON-BOARD SOLUTION | SMART SCHOOL BUS SOLUTION



STUDENT LEFT OVER CHECK

Preventing students from being left behind by combining driver - initiated checks, motion detection via PIR sensors, and alarm - triggering mechanisms to ensure student safety at the end of the journey.

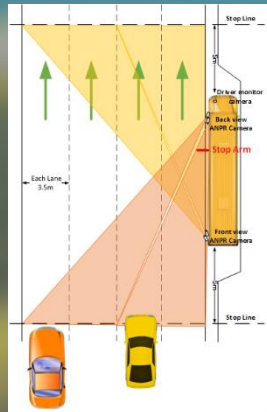
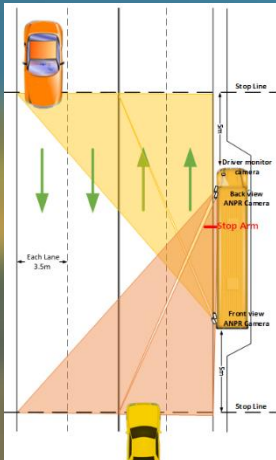
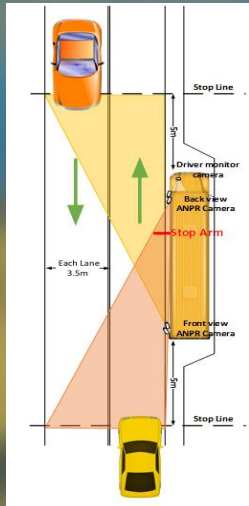


ON-BOARD SOLUTION | SMART SCHOOL BUS SOLUTION

STOP ARM SOLUTION

Enhancing student safety and reducing accidents via law enforcement and AI technology, while also generating revenue through imposing violation fines.

Scenario



When the STOP arm is opened, the movement of all vehicles is illegal ;



The license plate of the moving vehicle will be captured and identified as evidence of the fine ;

2 lanes、 4 lanes scenario

FLEET MANAGEMENT

This solution improves fleet efficiency, security, and decision-making with real-time tracking, system integration, and geo-fencing for enhanced security.



How can we ensure fast and reliable tracking of all Trucks in real-time?

- Implement GPS tracking with high-frequency updates and a robust network for live data transmission.
- Enables immediate response to incidents, enhances fleet visibility, and improves management efficiency.

How can we optimize fleet operations using data analysis?

- Analyze vehicle usage, equipment performance, and mileage data to identify inefficiencies and maintenance needs.
- Helps in predictive maintenance, reduces downtime, and enhances overall fleet performance.

How to use geo-fencing for boundary alarms?

- Use geo-fencing technology to set boundaries and trigger alarms when vehicles exceed predefined areas.
- Increases security, prevents theft, and ensures vehicles are used within designated routes or areas.

REDUCE OPERATING COSTS

By leveraging fuel consumption monitoring, insurance claim reduction, and proactive maintenance management, operating costs are significantly reduced through improved efficiency and minimized unexpected expenses.



How can we reduce fuel consumption and improve driving behavior?

- Implement real-time monitoring of driving behavior (e.g., idling, route deviation) using ultrasonic sensors or CAN Bus data to issue early warnings.
- Promotes fuel-efficient driving, reduces fuel costs, and provides reports for further analysis and optimization.

How can we reduce insurance costs and prevent fraud?

- Use video evidence from on-board cameras to accurately record incidents and lower the risk of false claims.
- Reduces insurance premiums, decreases fraudulent claims, and lowers accident-related costs.

How to monitor vehicle health in real time?

- Utilize in-vehicle sensors and remote communication technology to monitor vehicle health in real-time, identifying potential issues before they occur.
- Enables early intervention, reduces downtime, and lowers repair costs by scheduling preventive maintenance.

ON-BOARD SOLUTION | Platform Ability

Manage drivers' driving time and behavior according to driving violations such as ADAS, DMS, speeding, sudden acceleration and deceleration. Construct a safety risk assessment profile for drivers, identify high-risk drivers, focus on dangerous and bad driving behaviors and habits.

Multi - source Data



Driver algorithm recognition
Driving duration

Driver's identity



Rapid acceleration
Line straddling...

Bad driving habits

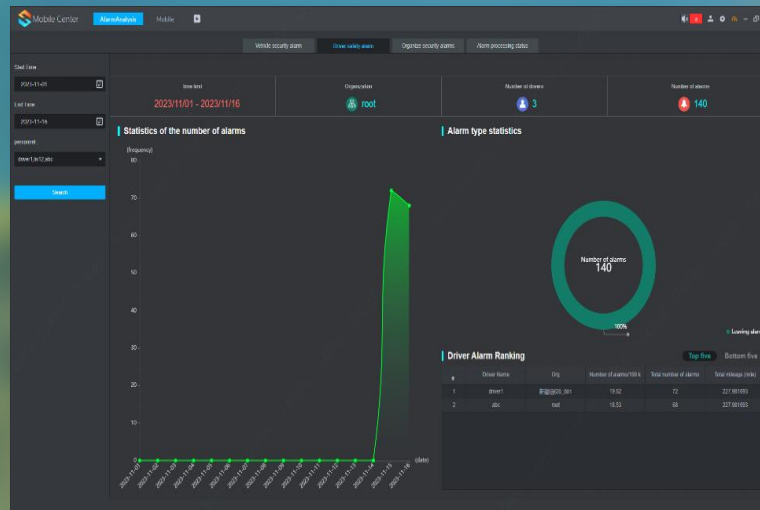


violate traffic regulations,
incident....

Violation driving records

Driving Safety Evaluation Model

Multiple Dimensions



Pursue liability after event

Take proactive preventive measures before event

Risk Control and Management



Identify high - risk
drivers



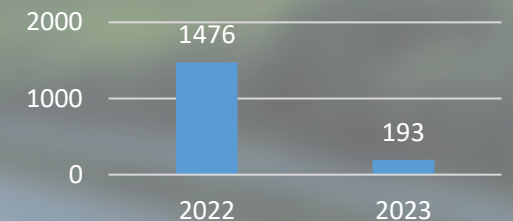
focus on dangerous
driving



closed - loop approach
in rectification

Solution Value

- Customer I owns 7,400 buses, and there are issues such as drivers' overtime driving and smoking or making phone calls while driving



NO. of Calling alarm

Smoking
89%
decrease

Calling
87%
decrease

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SUCCESS CASES | TURKEY BUS PROJECT



Project Background

The end user is the biggest bus company under the **Istanbul City Government**. The customer needs a mobile monitoring system with high compatibility and stability, and supports for expansion of the number of channels, **network sharing, intelligent management of camera's working hours, and a unified management platform**. Project budget is allocated by the city government.

Challenges

 Bad network connection and data sharing

 Too much accidents and economic losses

 Lack of scientific basis for route adjustment and bus scheduling

Project Video



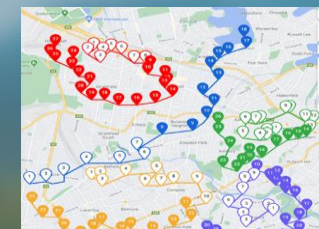
Project Highlight



Driver Status Monitoring



People Flow Monitoring



Plan Optimization

SMOKING

89%

CALLING

87%

Project Value

It shows that Dahua on-board solution are highly integrated, flexible and support in-depth development, and can meet the complex needs of various customers.



Strengthen Bus Safety



Improve Operational Efficiency



Improve Passenger Experience



Drive Sustainable Mobility

SUCCESS CASES | UAE SCHOOL BUS PROJECT



Project Background

In UAE ,The school bus is the main means of transportation for students to go to school. The School bus company **has more than 6000 school buses** and they transfer **more than 200k students** daily. This company has been cooperating with Dahua for 5 years and has been using Dahua's complete school bus solutions.

Challenges

- **Too much traffic accidents:** Accidents cause the economy losses, and the safety of students cannot be guaranteed
- **Students safety problem:** The students are too young and often fall asleep in the school bus without getting off. It's very dangerous in hot weather.

Project Highlight



Attendance Check



Violation Capture



Accident Prevention

- **Blind-spot visual assistance**
- **Left child detection**
- **Over speed/ hard brake/ rapid accelerate detection**
- **Get children location at any time**
- **Complaints hotline decreased by 80%**

Project Value



360° Monitor Reduces Accidents



Automatic Attendance Improve Efficiency



Student Left over Check, Prevent Student Safety

Project Background

PepsiCo, Inc. is an American multinational food, snack, and beverage corporation. It has many trucks delivering goods to all Egypt, during the transportation, many uncontrolled behavior need to be recorded.

Challenges

- **Too much traffic accidents :** Accidents cause for some of their trucks and losses of goods , and the safety of driver cannot be guaranteed
- **Track the truck root problem:** The truck not drive in same direction each time so they need to track the root of each driver with analysis for their behavior.

Project Value

- Dahua succeed in entering logistics market in Egypt and made a good reputation, Management can easily track all needed info of trucks, such as location, driver behavior, speeding status and also in the future, it will integrate with Dahua card reader to achieve attendance system.
- It shows that Dahua on-board solution are highly integrated, flexible and support in-depth development, and can meet the complex needs of various customers.

Project Highlight



Driver Behavior Monitoring



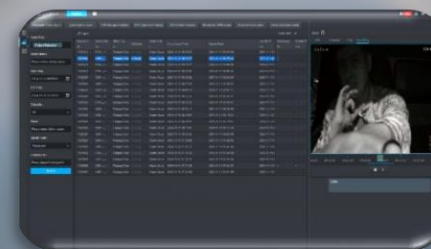
ADAS



DSM



MNVR



Mobility Integration Platform



Driving Behavior Regulation



Truck Location



Goods Trajectory